Knowledge Graph Completion

Subtitle

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Language Models and Structured Data

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Problem Statement

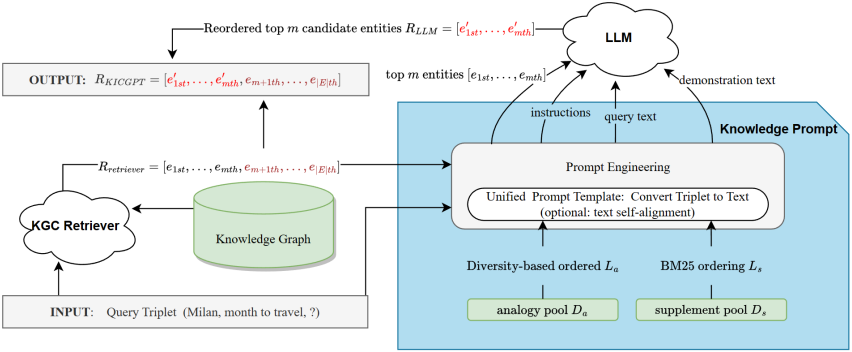
- Knowledge Graph Completion (KGC) aims to infer missing facts in a knowledge graph (KG).
- A KG is defined as $G = (E, R, T)$ where:
  - E is the set of entities
  - R is the set of relations
  - $T \subseteq E \times R \times E$ represents valid triples
- Two fundamental tasks:
  - **Link Prediction**: Predicting missing entities.
  - **Triple Classification**: Determining validity of a triple.

Background

- Early approaches used embedding-based models:
  - TransE, DistMult, ComplEx.
- Large Language Models (LLMs) improve KGC by leveraging:
  - In-context learning.
  - Instruction tuning.
- Two main strategies:
  - **Prompt-Based Methods** (e.g., KICGPT).
  - **Fine-Tuning-Based Methods** (e.g., KoPA).

KICGPT: Prompt-Based Link Prediction

- Uses retrieval and ranking for entity prediction.
- Steps:
  - Retrieve candidate entities.
  - Use Knowledge Prompts to refine ranking.
  - Final ranking by LLM-based reranking.



KoPA: Fine-Tuning-Based Triple Classification

- Uses Knowledge Prefix Adapter (KPA) to integrate KG structure.
- Embeddings guide LLM reasoning.
- Fine-tuned LLM predicts validity of triples.

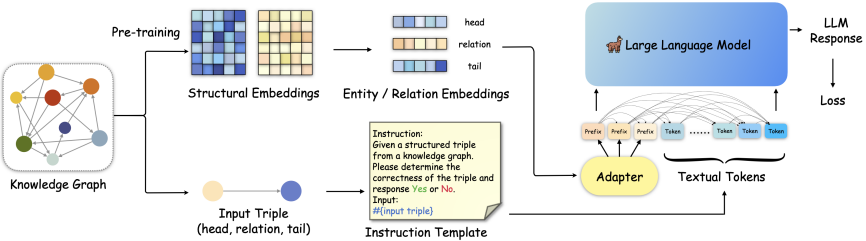


Figure: KoPA Architecture

Experiments

- Datasets: WN18RR and CoDeX-S.
- Evaluation metrics:
  - Mean Reciprocal Rank (MRR), Hits@1, Hits@3, Hits@10.
- Compared models: Alpaca-7B, Ministral-8B, R1-Distill-8B.

Dataset	# Entities	# Relations	# train	# valid	# test
WN18RR	40,943	11	86,835	3,034	3,134
CoDeX-S	2,034	42	32,888	3,654	3,654

Table: Statistics of the datasets used in our experiments.

Plots

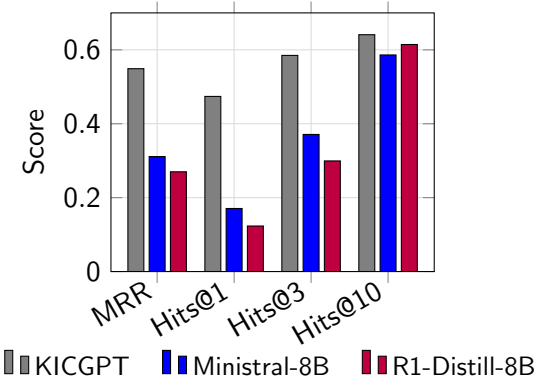


Figure: Link Ordering Evaluation Metrics

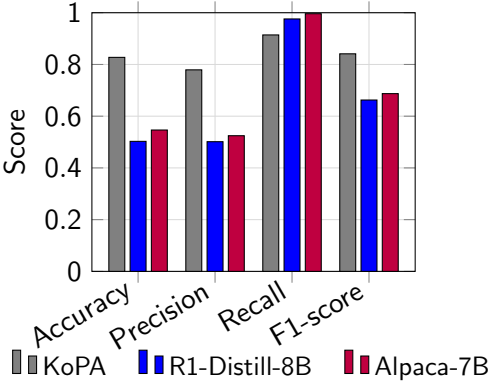


Figure: CoDeX Performance Evaluation Metrics

Plots

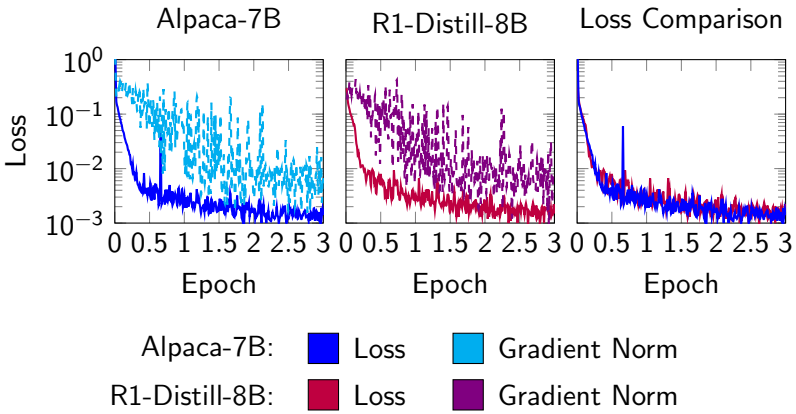


Figure: Fine-tuning of Alpaca-7B and R1-Distill-8B. Loss and gradient norm trends are shown separately for each model, along with a comparison of both losses. Figure and results from our work.

Discussion

- KICGPT improves ranking by leveraging structured prompts.
  - ChatGPT outperforms R1-Distill-8B and Ministral-8B due to better reranking heuristics.
- KoPA integrates KG embeddings effectively for classification.

Conclusion

TBD

Bullet Points

- Lorem ipsum dolor sit amet, consectetur adipiscing elit
- Aliquam blandit faucibus nisi, sit amet dapibus enim tempus eu
- Nulla commodo, erat quis gravida posuere, elit lacus lobortis est, quis porttitor odio mauris at libero
- Nam cursus est eget velit posuere pellentesque
- Vestibulum faucibus velit a augue condimentum quis convallis nulla gravida

Blocks of Highlighted Text

In this slide, some important text will be highlighted because it's important. Please, don't abuse it.

Block

Sample text

Alertblock

Sample text in red box

Examples

Sample text in green box. The title of the block is "Examples".

Multiple Columns

Heading

1. Statement
2. Explanation
3. Example

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer lectus nisl, ultricies in feugiat rutrum, porttitor sit amet augue. Aliquam ut tortor mauris. Sed volutpat ante purus, quis accumsan dolor.

Theorem

Theorem (Mass–energy equivalence)

$E = mc^2$

Citation

An example of the `\cite` command to cite within the presentation:

This statement requires citation [Smith, 2012].

References

Smith, J. (2012).
Title of the publication.
Journal Name, 12(3):45–678.